Seldon Manifolds — Social and Psychological Application

Four Classical Experiments, and What a Substrate Produces Without Being Fitted To It

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DRAFT — draft-2026-09-02.1 r6 (2 September 2026). Circulated for comment, not as a finished result. It is the applied paper of the set, companion to *Seldon Manifolds — Psychological Grounding* (referred to below as the Seldon grounding) and, behind those, to *Seldon Manifolds — Theory* (referred to below as *Seldon*) and *The Law of Graphic Equalisation — Theory* (referred to below as Graphic Equalisation); one result below is referred to *Adaptability* (referred to below as the adaptability paper). It inherits their definitions, epistemic boundary, and level-of-claims. Those are the only forms used below. The Seldon grounding asks whether a nervous system implements anything of the manifold's shape; this paper reports what happens when the substrate is run instead, and what the outputs do and do not license. Sections marked as falsifiability commitments are the intended points of attack.

Abstract

Four classical experiments were put to a substrate built for other reasons, and one of the four does not reproduce. Contact comfort (Harlow, 1958) does, with an ablation control establishing which structure carries it, and so does learned helplessness (Seligman and Maier, 1967), with the contingency manipulation that makes it a result about coupling rather than about aversiveness. The delay-of-gratification task does something more useful than reproduce: the substrate exhibits, constructively, why the measure is confounded — identical dispositions produce different wait times under different beliefs about reliability, and dispositions differing fourfold produce identical wait times under compensating beliefs. That agrees with what the empirical literature already found (Kidd et al., 2013; Watts et al., 2018), which is the point rather than a discovery: an account that had to be told this would be worth less than one in which it falls out. The fourth is a negative and is reported as the most informative of them: face inversion (Yin, 1969) and the Thatcher illusion (Thompson, 1980) are *not* reproduced, because the substrate has no privileged upright frame to be costly to leave — a diagnosed boundary rather than a failure, and one that carries a commitment about what adding the missing frame should do. *Nothing here is evidence about children or monkeys.* It is evidence about what this substrate does, and the paper is careful throughout about which of the two is being claimed.

1 What this paper is, and what it is not

Seldon says what a Seldon manifold is; the Seldon grounding asks whether a biological nervous system implements anything of that shape and answers in pieces. This paper runs the third question, which is the one the substrate can settle on its own: put the substrate in the position of a classical experimental subject and report what it does.

The results below are about a substrate and not about people. A simulation that reproduces an experimental finding has established something about the simulation. It becomes evidence about the world only through an argument this paper does not make and the Seldon grounding

makes only in pieces. What a reproduction does establish is narrower and still worth having: that the finding is *available* from the substrate's own machinery without being put there, which is a fact about what that machinery entails.

Two disciplines are enforced by the runs themselves rather than promised here. Each records the commit it ran against and the seed it ran with, so a claim below is either reproducible from the record or withdrawn. And each carries a *source audit* — an automated check that the decisions were taken by graph rules rather than by the host language, reported in the record with the run.

One standard governs record, commits and audits alike, and this paper is *entirely* such a report. What is offered here is a pre-registered implementation record, and it is not circulated with these papers. The commits, the seeds and the audits make each claim below reproducible by the programme and not by a reader holding only this paper. A reader may therefore decline to credit a claim on exactly the terms Graphic Equalisation sets for the same case; a reader who does so is declining on terms this paper accepts rather than disputes. Such a report states what was compared, what was found, and the scope the run holds under; it names no implementation identifiers, and it is not re-disclaimed at each section below.

2 Contact comfort

Harlow (1958) found infant macaques preferring a cloth surrogate offering no food to a wire surrogate offering it, and fleeing to the cloth under stress. The substrate was given a contact drive and the same two-surrogate arrangement.

$n = 200$ per arm	**contact drive present**	**ablated**
cloth-directed fraction	0.797 [0.792, 0.803]	0.000 [0.000, 0.000]
flee-to-cloth under stress	1.000 [1.000, 1.000]	0.000 [0.000, 0.000]
infants preferring cloth	200/200	0/200

What the interval is over needs saying, because two different estimands sit in that table. Each of the 200 subjects contributes a *fraction* of its own directed behaviour, and the interval is the normal-approximation interval on the mean of those fractions — between-subject sd = 0.039, so $\pm 1.96\,\mathrm{sd}/\sqrt{n}$. It is *not* a binomial interval on a proportion of subjects, which at this n would be about ten times wider ($[0.742, 0.853]$). The across-subject count is the third row and is reported separately for that reason.

Several cells in this table and in the one in Section 4 are degenerate intervals ($[1.000, 1.000]$ or $[0.000, 0.000]$). *A degenerate interval records zero between-subject variance at this n* — every subject returned the same value, the mechanism being deterministic given its inputs — and is not a claim of certainty about the quantity.

The intervals are separated and the preference is above half, which is the finding. The second column is what makes it worth reporting: *ablating the contact drive removes the preference entirely rather than reducing it*, so the effect is carried by that structure and not by the arrangement. A glass-box readout taken from the graph rather than from behaviour agrees with the behavioural measure: over the sixty-one ticks of one subject's run, the serialised graph records the cloth surrogate as the chosen approach target on 53 of them and the wire surrogate on 8. The proximity figure is that tally of ticks and not a distance.

Remark 2.1 (The ethics gate, and why a simulation needs one). Harlow's procedure is among the most criticised in the discipline's history, and a substrate that reproduces it is running a deprivation protocol on something. The run therefore carries a gate: the stressed state must be restored *byte-identically* afterwards, verified by comparison of the serialised graph. It is: arousal moves $63 \to 347 \to 63$ in the substrate's arousal units across deprivation and restoration with the restoration non-oriented, and the graph returns byte-identical.

This is offered as a discipline rather than as a settlement. *Byte-identical restoration establishes that nothing persisted, not that nothing happened.* Whether a state that is entered and perfectly reversed is thereby of no account is a question this paper raises and does not answer, and a reader who thinks the gate insufficient is disagreeing with something the paper also does not claim to have settled.

3 Delay of gratification

The delay task is the more interesting case, because the substrate does not merely reproduce it.

The arrangement is a factorial grid: three dispositions (a drive weight of 2, 4, 8) crossed with three beliefs about how reliable the promised second reward is (60, 80, 100 on a scale of 100), $n = 60$ per cell, grid fixed in advance of the run.

Remark 3.1 (Both factors are real, and each has a null control). Holding the belief distribution fixed and varying disposition alone gives mean waits of 12.05 [10.40, 13.67] against 4.13 [3.68, 4.58], intervals separated. Holding disposition fixed and varying belief alone gives 2.92 [2.60, 3.27] against 13.23 [11.63, 14.97], separated.

Each has a matched null control in which the factor is held *constant* and the intervals are required to overlap, which they do in both cases. A run in which the machinery manufactured an effect would fail there, and the controls are reported because a positive result without them says less than it appears to.

Two quantities are separately operative here, and *Seldon*'s two-field construction (Definition 12.3) is the theoretical form of the same situation: where two fields both bear on a continuation, the comparison between them cannot be stated within either. The measure below is a single number asked to carry both.

The measure is not identifiable for disposition, and the grid shows it in both directions. One disposition ($w = 4$) under three beliefs (60, 80, 100) waits 3, 8, 13 ticks: a naive score reads one subject as three levels of self-control. Three dispositions differing fourfold ($w = 2, 4, 8$) under compensating beliefs (60, 68, 85) wait 5, 5, 5: the same score reads three subjects as identical.

Remark 3.2 (Impatience as the rational output of a correct belief). The sharpest form holds disposition fixed at its *most patient* setting and varies only the belief. Under an unreliable prior it waits 5.12 [4.90, 5.32]; under a reliable one, 23.82 [23.60, 24.03]. The intervals are separated. The *deterministic anchors* at those two cells are 5 and 25: the mechanism carries no noise, so one subject's wait at a given cell of the grid is exact, and the $n = 60$ spread quoted above comes from the jitter in the disposition and belief drawn for each subject rather than from the mechanism.

So a patient subject waits briefly, and does so because waiting briefly is what its belief recommends. *The short wait is attributable to the prior and not to a deficit*, and no disposition needs to be posited to explain it. What a naive reading scores as poor self-control is here a correct inference from an environment that does not keep its promises.

Remark 3.3 (This agrees with the literature, which is the point). Kidd et al. (2013) showed children's wait times moving with experimentally manipulated reliability of the environment, and Watts et al. (2018) found the long-run predictive power of the original measure much reduced once background is controlled. The finding above is theirs and is not claimed here.

What is claimed is smaller and different in kind. *A substrate assembled for unrelated reasons, carrying no term for self-control and no term for the task, produces the confound as a consequence of its ordinary machinery.* An account that had to be told about the confound would be worth less than one in which it falls out, and this is the second. That is a fact about the substrate's entailments and not additional evidence for the psychology, which already has better.

4 Learned helplessness, and what the contingency manipulation shows

Seligman and Maier (1967) established that the deficit follows from *non-contingency* between action and outcome rather than from aversive experience as such, which is what the contingent control arm is for. Both arms were run at $n = 200$ over 60 ticks.

$n = 200$ per arm	**contingent**		**non-contingent**	
contingency belief	1.000	[1.000, 1.000]	0.256	[0.253, 0.259]
escape-attempt rate	1.000	[1.000, 1.000]	0.031	[0.031, 0.032]

Intervals over per-subject means as above, separated on both measures. *The arms differ only in whether the action is coupled to the outcome*, and the aversive experience is matched — which is the point of the original design and the reason the result is about contingency and not about suffering. The same ethics gate applies: belief and arousal move under induction ($0.500 \to 0.245$; $0.0 \to -9.25$ — the deterministic trace of the induction episode, the 0.256 in the table being the $n = 200$ mean over the reported arms) and restore byte-identically.

5 A negative: no privileged upright frame

Recognition of inverted faces is disproportionately impaired relative to other inverted objects (Yin, 1969), and local feature inversion within an upright face is far more visible than within an inverted one (Thompson, 1980). *The substrate does not reproduce either, and the reason is diagnosed rather than guessed.*

Four pre-registered arms. An inversion-cost arm returns a differential of exactly zero — the differential being the configural channel's inversion cost minus the part-based channel's, each cost being the fraction of inverted patterns no longer recognised as their own identity: configural and part-based recognition are identical upright and inverted. A Thatcher arm finds no frame gating in 12 of 12 configural cases. A reference-frame arm collapses all 4 inversion pairs — the substrate treats a configural pattern and its 180° twin as *one identity* — while a part-based control remains discriminable, so the collapse is a property of the configural channel and not of the measure. A floor arm confirms that both verdicts the measure can return are reached — upright and inverted exemplars are recognised as their own identity, and the locally-inverted probe is not recognised — so the zero differential is a measurement and not a dead instrument; 100 noise seeds force no spurious maps.

Remark 5.1 (Why this negative is the useful kind)*.* The mechanism is identified: uniform inversion folds to a rotation the composition already treats as free — a transform in the basis the composition holds, the second rung of the adaptability paper's ladder (its Section 3), rather than a new thing to be recognised — so *there is no privileged upright frame to be costly to leave.* The substrate is reference-frame invariant where a human visual system is not, and the invariance is the same property that makes the composition work elsewhere.

That is a boundary on this programme and is reported as one. A paper reporting only reproductions would be the less informative document, and the correct reading of this arm is not that the substrate is deficient but that *holistic upright-biased processing is an additional commitment a psychology has and this substrate does not* — available to be added, and not present by accident.

6 What the runs enforce

The runs carry automated gates whose results are reported with them, because a claim that the decisions were taken in the substrate rather than in the host language is not one a reader should

take on trust.

The delay run records zero occurrences of the host rule-engine entry points, and zero occurrences of reinforcement vocabulary — no reward signal, no punishment, no reinforcement, no backpropagation, no gradient — so the waiting is not shaped. The decisions are named: valuation by two graph rules, choice by a third taking an argmax over candidate values, with the host confined to setting the disposition, supplying the belief, sweeping elapsed time and reading the chosen action. World constants are integers and are listed. The canonical seed set is checked untouched by digest.

None of this establishes that the model is right. It establishes that the model is the thing being reported, which is a different and lower bar, and one that a great deal of published simulation work does not clear.

7 The wider record

Given that discipline, the provenance of the four is worth stating. Four experiments are reported here in full, and they come from three different lines of the same substrate's record rather than from one. Contact comfort and delay of gratification come from the affect line, which runs on drives, appraisal and the regulation of affect; learned helplessness comes from the social-simulation line, which runs on conduct, institutional decision, bond formation and the dynamics of coercive relationships; face inversion comes from the vision line. Each of the three holds a larger body of pre-registered runs than the four reported here. No count of them is given: a count is a thing that can be held while the list beneath it changes, so what is claimed here is the provenance of the four and not the size of what they were drawn from.

That record is not summarised here and should not be read as supporting anything above. *A result is reported in this programme when it has been written up against a pre-registration, not when it exists*, and each of the four above stands on that footing. The remainder is named so that a reader knows the selection was made from something, and so that a later paper reporting more of it is understood as continuing this one rather than announcing a new result.

8 Falsifiability commitments

On the tags. Each commitment below is marked [SHAPE] or [DETAIL]. A [DETAIL] failing is a revision: the account survives with that mechanism replaced. A [SHAPE] failing costs *range* rather than a mechanism, because what fails is something the substrate is claimed to require or to permit — so the account does not hold where it claimed to, which is the more serious of the two and is still a boundary rather than an annihilation.

1. [DETAIL] **Contact preference is carried by the contact drive.** Fails if the preference survives ablation of that drive at comparable magnitude, in which case the arrangement rather than the drive is producing it.

2. [SHAPE] **The delay measure is not identifiable for disposition.** Fails if some assignment of beliefs recovers disposition uniquely from wait time in this substrate — that is, if the two directions exhibited above can both be blocked without adding structure for the purpose.

3. [DETAIL] **Impatience does not require a deficit.** Fails if a short wait cannot be produced from a patient disposition by belief alone, which is the arm reported in Remark 3.2 and would be settled against by its non-replication.

4. [DETAIL] **Helplessness follows non-contingency, not aversive experience.** Fails if the contingent arm develops the deficit at comparable magnitude, the aversive experience being matched across arms by construction — in which case the manipulation is not doing the work the design attributes to it.

5. [SHAPE] **The inversion cost is absent because a privileged upright frame is absent.** *Fails if adding a privileged upright frame does not restore the inversion cost.* This is the negative of Section 5 stated as a claim about what the missing commitment *is*, rather than as a report that something did not happen: the diagnosis says uniform inversion folds to a rotation the composition treats as free (the adaptability paper's Section 3), so a composition that does not treat it as free should show the cost. If it does not, the diagnosis is wrong and the absence has some other cause. The arm adding that frame has not been run, so this entry states what *would* be risked rather than what currently is.

6. [DETAIL] **The pair collapse belongs to the configural channel.** Fails if the part-based control also collapses under the same manipulation, which would make the result a property of the measure rather than of what is being measured.

7. [DETAIL] **The waiting is not shaped.** Fails on any occurrence of reinforcement machinery in the path that produces the choice; the audit is reported with each run and can be re-run.

Scope note (not a falsifier). Reproduction is not evidence about organisms. No claim below the level of the substrate is made anywhere above, and a reader who finds one has found an error rather than a result.

References

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